1. **Core loop(Caleb)** in one sentence:

The player uses Energy Blasts to defeat enemies and get XP, which they can then use to upgrade their skill tree, thus improving their abilities/movement.

2. **Mechanics(Aiden, Rhaiyn, Caleb):** At least two per member. For each: its name, its owner, the input that does it, the rules, and the numbers.

Caleb:

1. Energy Blast
    a. Input: Left mouse button
    b. Rules: Each energy projectile deals 20 damage to enemies, costs 10 energy, and has a 0.5-second cooldown
2. Energy Recharge
    a. Input: Hold “R”
    b. Rules: Player restores 15 energy per second while holding input, up to a maximum of 100. While recharging, movement speed decreases by 30%

Aiden:

1. Skill Tree
    a. Input: k (can also reach it from the menu)
    b. Rules: Skill Tree affects dodge by giving it extra abilities like a damage increase on a perfect dodge and the ability to chain two dodges.
2. Dodge
    a. Input: Shift(Remapable)
    b. Rules: The player performs a 0.4-second dash in the direction they're moving. It gives invulnerability with a 0.5-second dash cooldown. A perfect dodge is done within 0.133 seconds of an attack hitting.

Rhaiyn:

1. Grapple
    a. Input: Q
    b. Rules: The player grapples to a point within 12 meters and is pulled toward it at 14 meters per second. Grapple has a 4-second cooldown. After the grapple ends, the player gains Momentum for 1.5 seconds

2. Ground Slam
    a. Input E (While Airborne)
    b. Rules: The player slams toward the ground and deals 30 damage to enemies within a 3-meter radius. Ground Slam has a 5-second cooldown. If used during Momentum, it deals 50 damage within a 5-meter radius

3. **How they plug in(Aiden, Rhaiyn, Caleb):** For each member, which of their own mechanics each one changes, and which teammate's mechanic one of them changes, and how.

Energy Blast (Caleb): Energy Blast changes Energy Recharge because successfully hitting an enemy restores 5 energy, allowing the player to recharge through combat.
Energy Recharge (Caleb): Recharge changes Energy Blast because reaching at least 75 energy increases the next blast's damage from 20 to 30.

Perfect Dodge (Aiden): Perfect dodging doubles the currency/xp gained for 5 seconds, making it easier to gain new skill points faster. Perfect dodging again during the 5 seconds resets the 5-second timer.

Skill Tree (Aiden): Skill Tree affects dodge by giving it extra abilities like a damage increase on a perfect dodge and the ability to chain two dodges.

Grapple (Rhaiyn): Grapple changes Ground Slam because Grapple gives the player Momentum, which increases Ground Slam's damage from 30 to 50 and its radius from 3 meters to 5 meters.

Ground Slam (Rhaiyn): Ground Slam changes Grapple because hitting at least one enemy with Ground Slam reduces Grapple's remaining cooldown by 2 seconds.

4. **Win and lose(Caleb)**

   The player wins by defeating each Wave room. The player loses if they die.

5. **Art direction (Aiden, Rhaiyn, Caleb):**
   a. **The world, in one sentence.**

      A steampunk city with a massive office skyscraper with over 100 floors (exaggeration) in the center.

   b. **Palette(Aiden):** Five colors with hex values. One is the accent, and it appears in almost nothing.
      i. #887972
      ii. #e1c289
      iii. #a7793e
      iv. #6b4131
      v. #856045
   c. **Camera and framing(Caleb, Aiden):** What angle is everything viewed from, and at what apparent scale?

      Top-down, slightly angled isometric perspective. The camera follows the player while keeping the surrounding combat arena visible. Enemies, projectiles, and environmental hazards should be visible within the camera's viewport.

d. **Light(Caleb):** Direction, hardness, temperature, and whether there is any fill.

   The play area is illuminated by default regardless of light sources, so the player can see at all times. Temperatures are warm, but the environment itself is a hard/harsh industrial scene.

e. **Edge and detail(Aiden):** How does an object meet the background, and where is detail allowed to live?

   The player is the central subject with high detail and a sharp image. The background lives in the background walls, living in the foreground. The player lives in the middle ground.

f. **Mood, in three words(Caleb):** Three, not five. Specific enough that two of your classmates could not have written the same three.

   Industrial, Steam, Machine

g. Style Suffix(Rhyain):

   Top-down perspective, slightly angled isometric perspective. Muted warm industrial palette: #887972, #e1c289, #a7793e, #6b4131, #856045; use one color as an extremely rare accent. Constantly illuminated, fully visible play area regardless of light sources. Warm color temperature contrasted with harsh industrial environments. Clear foreground, middle-ground, and background separation: foreground contains framing industrial structures, middle ground contains the highly detailed, sharp, central player and enemies, background contains distant walls and machinery with reduced detail and contrast. Never obstruct gameplay visibility. Industrial, Steampunk, Mechanical aesthetic.

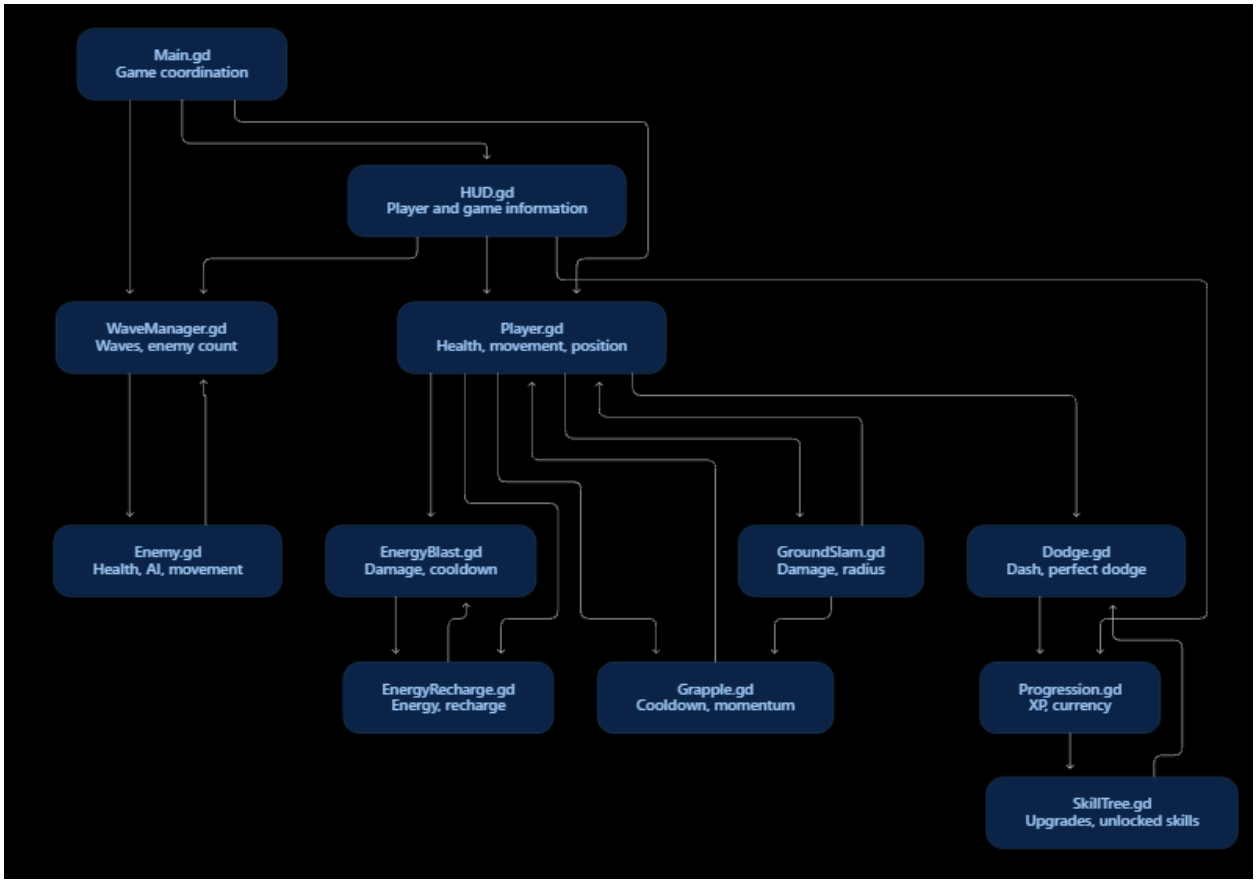
h. Avoid List(Caleb):

   Chrome, bright blue, bright colors in general.

i. Anchor Asset:

6. **Architecture(Aiden):** A diagram, drawn before you build: your scenes and scripts, where each mechanic's state lives, and which script reads which. Boxes and arrows are enough: one box per scene or script, labeled with the mechanic state it holds, and an arrow from each script to every script it reads. Your code

will differ, and that costs nothing here.



7. **Scope and cuts(Aiden, Caleb):**

   Scope: Player movement, HUD, the 6 mechanics, and bosses

   Cut:

   - The presumed “futuristic” art style and design we had settled on felt too predictable and overdone, so we changed the game's theme to a steampunk-esque top-down wave survival shooter.
   - Procedurally generated skyscraper floors.
   - Destructible environments.
   - Multiplayer or online functionality.